

on its back. But wait for prices to fall or a discount sale to trigger because as of now, buying a solar power bank just doesn't make a lot of sense.

However, buying solar-powered LED lights is definitely a good idea. Check the ones sold on Amazon (like this one – <http://dg.it.in/SwftlyDnLED>) that cost around ₹1k per unit. They are remarkably easy to install and glue on to any substance fairly quickly, whether it's indoors or outdoors. Charge through the day and activate at night. What's more, they are only activated when they sense motion. Pretty effective and efficient, huh?

## MAKE DUMB TVs SMARTER FOR PENNIES

You might cringe every time you look at full page advertisements on the front page of your daily newspaper splashing across TVs costing tens of thousands and even lakhs of rupees. Touting “smart” features such as access to an app ecosystem, Web browsing and screen-casting technology, you don't really need to spend all that money just to get a “Smart TV”.



Remember, just like a camera is all about optics and the ability to take great photos, so it is with a TV – considering the type of panel, refresh rate and other core display specifications. Everything else, especially the so called smart features can be found elsewhere. As long as your HDTV has an HDMI port, you can make it smart. All these jargons and more have been covered in our FT to Display Technology, so be sure to check it out.

If you're a tinkerer at heart, look no further than the Raspberry Pi 2 which can be tweaked with appropriate software in order to convert it into a media streaming device. You then connect it to your TV using an HDMI cable and voila! You have your very own, very cheap, Smart TV alternative. If you want a little more out of the box setup and user experience, try checking out the new Chromecast. It connects to any HDMI-enabled TV and allows you to

stream content from your phone or laptop on to the TV. It also comes with built-in app support to ensure the whole process is as seamless as possible. The Chromecast sells for about ₹3k.

Beyond Chromecast, there's Teewe 2 which is a nifty little HDMI dongle that has been made by Indians, in India. Like the Chromecast, the Teewe 2 is a streaming stick available on most e-commerce websites, at ₹2,399.

So yeah, don't be tempted to buy that next best Smart TV only for its smart features. Because you'd be doing your bank balance and your sense of stingy ingenuity a whole lot of disservice.

## TAKE YOUR CAR'S IQ FROM LOW TO HIGH

The TV analogy given above also applies to cars and other automobiles. If you don't want to spend on buying the latest Mercedes or Audi or BMW for its style statement and comfort of ride on the road or elsewhere, then you're better off scavenging some of these high-end luxury vehicles' “smart” features through inexpensive alternatives – not cheap knockoffs. By doing this, you will make your old, faithful automobile smarter and just as useful in the realm of gizmo wizardry as the high-end European cars.

Actually, whether it's brand new or old as your office furniture, your car is smarter than you think. If your car's built after 1996, and you've been driving it since then, you can access a treasure trove of data inside it which goes beyond just the speedometer. Purchase a nifty little Bluetooth dongle called Automatic (US \$100), to start communicating with your car like never before. It automatically plugs into any car's standard diagnostic (OBD-II) port. It accesses and unlocks the data in your car's on-board computer and connects it to your phone wirelessly using Bluetooth.

You are going to get a whole lot of info from your car, so be prepared. Telemetry data, track mileage, car location, and many other features incoming with the Automatic adapter and app sync.

Similar products that are launched and made in India

include CarIQ and Zene. Both products offer similar solutions like the Automatic plug, with the added bonus of being customised and made specifically for the Indian market to support cars sold in India. Both the products are available for purchase on their respective websites.

## BREATHE NEW LIFE INTO OLD HARDWARE

Just because your gadgets are old and not as fast or effective as they were in their prime doesn't mean you have to get rid of them. Far from it. Here's what you can do from old, ailing hardware like your old PC or laptop.

- **Convert it to a NAS:** Install FreeNAS on your old PC. After doing this and sharing folders with data on it, and obviously connecting it to the network, you can use your old PC as a network drive.
- **Retro console:** Install FreeDOS or any other standard NES emulator and turn your old PC into a retro gaming console. For free.
- **Jukebox on request:** Buy a decent pair of speakers (like the F&D F680s), plug them into your old PC, stick in your music or connect it to Spotify, Gaana or Apple Music and forget about buying a dedicated music player.
- **Hotspot:** If you have an aging laptop that you're thinking of getting rid off, stop, don't do it just yet. You can use it as a Wi-Fi hotspot or Network Repeater if your house isn't fully getting covered by your wireless router's signal.
- **Firewall:** Connecting to the Internet can be a security risk. In these cases, you can convert your old PC as a hardware firewall to ensure all incoming traffic

is routed through it and it's the first defence against the Web. You can also apply parental controls. Surf the Web for open source tools, there are plenty of them around.

- **Cloud storage:** If you don't want to submit your data on a third-party cloud service, install something like ownCloud on your old PC and an external drive to host your own always-on private cloud which will be accessible from anywhere on the web. 

